

Statistical Inference & Hypothesis Testing Report

Analyst: Portfolio Toolkit | Date: June 02, 2026 | Framework: AP Statistics Standard

1. State

Define the hypotheses and significance level to evaluate the claims of the A/B test.

Significance Level (α): 0.05

Hypotheses (Symbols):

$$H_0 : p_A = p_B \quad H_a : p_A \neq p_B$$

Null Hypothesis (H_0): There is no difference in the conversion rates between Group A and Group B ($p_A = p_B$).
Alternative Hypothesis (H_a): There is a statistically significant difference in the conversion rates between Group A and Group B ($p_A \neq p_B$).

2. Plan

Identify the appropriate statistical procedure and verify its mathematical assumptions.

Recommended Procedure: Two-Sample Z-Test for Proportions

Assumption / Condition Checked	Status & Diagnostic Details
Randomness	Assuming samples are collected via independent random selection.
Independence	Samples represent less than 10% of their respective target populations.
Success-Failure Condition	Group A successes (11654) & failures (84071) ≥ 10 ? Yes . Group B successes (11264) & failures (84424) ≥ 10 ? Yes .

■ CONDITIONS MET

All mathematical conditions for the Two-Sample Z-Test for Proportions are satisfied. Parametric results are highly reliable.

3. Do

Calculate sample metrics, standard error, test statistic, and the resulting p-value.

Sample Descriptive Statistics:

Group	Sample Size (n)	Successes (x)	Conversion Rate (p)
Group A	95725	11654	12.1745%
Group B	95688	11264	11.7716%

Calculations & Mathematical Formulations:

- Pooled Prop:

$$p_{pooled} = \frac{x_A + x_B}{n_A + n_B} = 0.1197$$

- Se:

$$SE = \sqrt{p_{pooled}(1 - p_{pooled})(\frac{1}{n_A} + \frac{1}{n_B})} = 0.0015$$

- Z Stat:

$$Z = \frac{\hat{p}_B - \hat{p}_A}{SE} = -2.7146$$

- Ci:

$$CI = (\hat{p}_B - \hat{p}_A) \pm Z_{crit} \times SE_{diff} = (-0.0069, -0.0011)$$

Statistical Parameter	Value
Test Statistic (Z)	-2.71461
p-value	0.00664
95% Confidence Interval	(-0.0069, -0.0011)

4. Conclude

State the final decision and offer a business interpretation based on the statistical results.

Compare p-value to α : $0.00664 < 0.05$. Since the p-value is less than the significance level, we **reject the null hypothesis (H_0)**. There is strong evidence supporting a statistically significant difference between the two variations.

■ DECISION: REJECT NULL HYPOTHESIS

Business Action Recommendation: The conversion rate difference of 0.40% is statistically significant. Deploy **Group A** to production, as it is expected to generate a permanent uplift in performance.